

Session 10

1. Create a React Context called useContext in a new file useContext.js and provide a default value with a username and a loggedIn status.
2. Build a simple Navbar component that uses useContext to display the current username from useContext.
3. Implement a toggle button in your app that switches between 'light' and 'dark' themes using Context API, and update the background color of the main div accordingly.
4. Refactor an existing component that passes theme as a prop through multiple levels; instead, use useContext to access the theme directly in the deepest child component.

 Hint: Remove all theme-related props and rely only on the context value.
5. Use ChatGPT or GitHub Copilot to generate a code snippet for a React context that manages notification count (like unread messages in WhatsApp), then integrate it into a small demo component that displays the count.

Answer:-

UseContext.jsx

```
import { createContext, useState } from "react";
import ThemeContext from "../ThemeContext";

export const themeContext = createContext();

const UseContext = () => {
  const [theme, setTheme] = useState("light");

  const handleTheme = (e) => {
    if (e.target.checked) {
      setTheme("dark");
    } else {
      setTheme("light");
    }
  }

  return (
    <themeContext.Provider value={theme}>
      <ThemeContext />
      <input type="checkbox" onChange={handleTheme}>
    </themeContext.Provider>
  );
}
```

```

        /> {theme}
      </themeContext.Provider>
    )
  }
}

export default UseContext

```

ThemeContext.jsx

```

import { useContext } from "react";
import './style.css';
import { themeContext } from './UseContext';

const ThemeContext = () => {
  const theme = useContext(themeContext);
  return (
    <div className={`panel-${theme}`}>
      <p>Hello guys</p>
      <p>kaha ho bhai</p>
      <p>maje kro</p>
    </div>
  )
}

export default ThemeContext

```

NotificationContext.jsx

```

import { createContext, useState } from 'react';
import NotificationDemo from './NotificationDemo';

export const NotificationContext = createContext();

const NotificationProvider = () => {
  const [count, setCount] = useState(0);

  const addNotification = () => {
    console.log(count);
    setCount(count + 1);
  }
}

```

```

const removeNotification = () => {
  console.log(count);
  setCount(0);
}

return (
  <NotificationContext.Provider value={{ count, addNotification,
removeNotification }}>
    <NotificationDemo />
  </NotificationContext.Provider>
)
}

export default NotificationProvider

```

NotificationDemo.jsx

```

import { useContext } from 'react';
import { NotificationContext } from '../NotificationContext';

const NotificationDemo = () => {
  const {count, addNotification, removeNotification} =
useContext(NotificationContext);
  return (
    <div className='flex gap-1.5 flex-col text-center'>
      <h3>Whatsapp Notification</h3>
      <button onClick={addNotification} className='border'>receive
New</button>
      <button onClick={removeNotification} className='border'>Clear All
message</button>
    </div>
  )
}

export default NotificationDemo;

```

App.jsx

```

import NotificationProvider from '../Session10/NotificationContext'
import UseContext from '../Session10/UseContext'

const App = () => {
  return (
    <div>

```

```
    <UseContext />
    <NotificationProvider />
  </div>
)
}

export default App
```